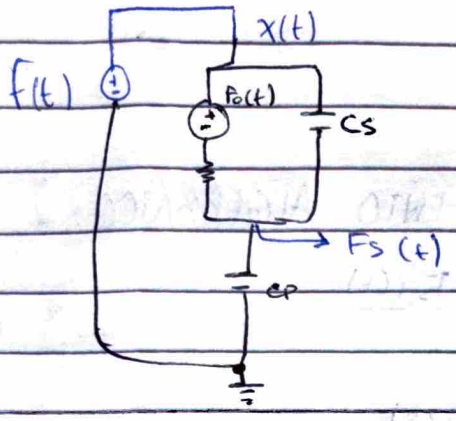
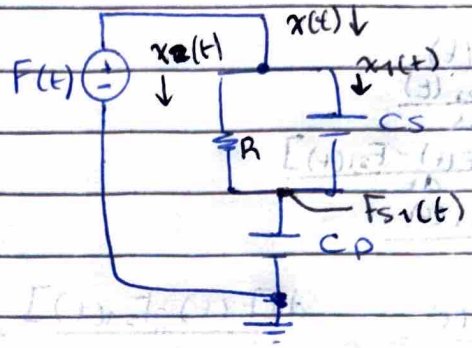
Práctica 55. Modelado y control del sistema musculoesquelético

la salida =

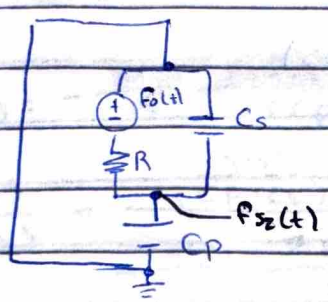
$F_s(t) = F_{s2}(t) + F_{s1}(t)$

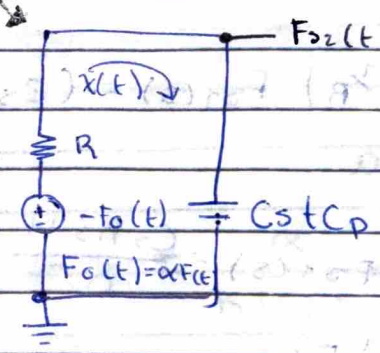
Circuito 1, por superposición

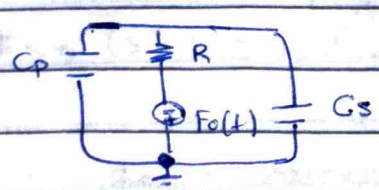




Circuito 2

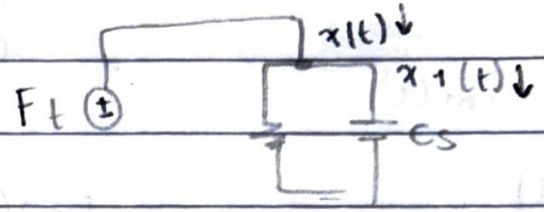






Análisis del circuito por superposición

- Circuito 1 : Apagando $F_0(t)$



$x(t) = x_1(t) + x_2(t)$

$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$

$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$

$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$

PROCEDIMIENTO ALGEBRÁICO

$$C_p = \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p S F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p S F_{s1}(s) = C_s S F(s) - C_s S F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$(C_p s + C_s S + 1/R) F_{s1}(s) = (C_s S + 1/R) F(s)$$

Suma algebraica

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

- Circuito 2: Apagando $F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$-\alpha F(s) = -\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{-\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\boxed{\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}}$$